

DISPROOF OF A CONJECTURE OF AKBARI, ELPHICK, KUMAR, PRAGADA AND TANG ON THE SIGNATURE OF LINE GRAPHS

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ABSTRACT. Let $n^+(G)$, $n^0(G)$ and $n^-(G)$ denote the numbers of positive, zero and negative eigenvalues of the adjacency matrix of a graph G , and let $s(G) = n^+(G) - n^-(G)$ be its signature. Akbari, Elphick, Kumar, Pragada and Tang [*Discrete Math.* **349** (2026) 114953, Conjecture 4.1] conjectured that every connected graph G satisfies $n^+(L(G)) \leq n^-(L(G)) + 1$, equivalently $s(L(G)) \leq 1$. We disprove this conjecture. We exhibit a connected graph on 14 vertices whose line graph has inertia $(9, 0, 7)$, and an infinite family F_k ($k \geq 4$) of connected graphs with $\text{In}(L(F_k)) = (3k + 1, 0, 3k - 1)$, so that $s(L(F_k)) = 2$. We further exhibit connected graphs with $s(L(G)) = 6$, showing that $s(L(G))$ is not bounded above by any absolute constant. All inertias are certified by exact rational arithmetic. We further show, by an exhaustive search over all 775,184,808 connected graphs in the relevant range on at most 12 vertices, that a smallest counterexample has order 13 or 14.

1. INTRODUCTION

All graphs here are finite, simple and undirected. For a graph G we write $n(G)$ for its order, $m(G)$ for its size, and $A(G)$ for its adjacency matrix. The *inertia* of G is the triple $\text{In}(G) = (n^+(G), n^0(G), n^-(G))$ recording the numbers of positive, zero and negative eigenvalues of $A(G)$, counted with multiplicity, and the *signature* of G is $s(G) = n^+(G) - n^-(G)$. The *line graph* $L(G)$ has the edges of G as its vertices, two being adjacent when they share an endpoint.

Akbari, Elphick, Kumar, Pragada and Tang [1] conjectured that $2n^+(G) \leq n^-(G)(n^-(G) + 1)$ for every graph G , a statement which would extend the absolute bound for strongly regular graphs to all graphs. In the course of verifying that conjecture for line graphs they were led to propose the following considerably sharper statement for line graphs specifically.

Conjecture 1.1 ([1, Conjecture 4.1]). For any connected graph G with line graph $L(G)$,

$$n^+(L(G)) \leq n^-(L(G)) + 1,$$

equivalently $s(L(G)) \leq 1$.

The evidence assembled in [1] for Conjecture 1.1 is substantial. The authors verify it for every graph on at most 9 vertices and for numerous graphs G with $m(G) \leq 100$ drawn from standard databases. They prove it for line graphs of trees, using a theorem of Guo [3] on the distribution of Laplacian eigenvalues of trees, and they prove it whenever $m \geq 2n - 1$ via the estimate

$$n^+(L(G)) \leq n \leq m - n + 1 \leq n^-(L(G)) + 1,$$

so that, as they observe, *Conjecture 1.1 is open only in the range $n \leq m \leq 2n - 2$* . They also record several families attaining equality: the cycles C_{4k+1} for $k \geq 1$, two disjoint copies of C_5 joined by an edge, and various kayak paddle, tadpole and prism graphs. Finally they note that connectivity cannot be dropped, since $C_5 \cup C_5$ is a disconnected counterexample.

Date: July 24, 2026.

2020 Mathematics Subject Classification. 05C50, 05C76.

Key words and phrases. inertia; signature; line graph; signless Laplacian; graph eigenvalues.

Remark 1.2. Conjecture 1.1 should not be confused with the main conjecture of [1], namely $2n^+(G) \leq n^-(G)(n^-(G) + 1)$, which Chen and Li [5] disproved (together with the weaker variant also proposed in [1]) by means of a family of reduced graphs W_k with $\text{In}(W_k) = \left(\binom{k}{2} + 1, 0, k - 1\right)$. Their counterexamples are not line graphs: the main conjecture of [1] was proved there for line graphs [1, Theorem 4.5] in the sharper form $n^+(L(G)) \leq \frac{1}{2}n^-(L(G))(n^-(L(G)) + 1)$, whereas W_5 has $n^+ = 11$ and $n^- = 4$, so that $11 > \frac{1}{2} \cdot 4 \cdot 5 = 10$. The two refutations are therefore independent. Conversely, the graphs constructed below satisfy the main conjecture of [1] with room to spare: for F_k as in Theorem 1.4 one has $2n^+(L(F_k)) = 6k + 2$ while $n^-(L(F_k))(n^-(L(F_k)) + 1) = 3k(3k - 1)$.

In this note we show that Conjecture 1.1 is false. Our counterexamples all lie in the range $n \leq m \leq 2n - 2$ identified in [1] as the only one left open.

Theorem 1.3. *Let G_0 be the graph obtained from a 4-cycle by attaching, at each of two adjacent vertices, a 5-cycle joined by a single edge. Then G_0 is connected with $n(G_0) = 14$ and $m(G_0) = 16$, and*

$$\text{In}(L(G_0)) = (9, 0, 7), \quad s(L(G_0)) = 2.$$

Theorem 1.4. *For $k \geq 1$ let F_k denote the connected graph on $5k + 1$ vertices obtained from k disjoint 5-cycles by joining one vertex of each to a common new vertex. Then for every $k \geq 4$,*

$$\text{In}(L(F_k)) = (3k + 1, 0, 3k - 1), \quad s(L(F_k)) = 2.$$

Moreover $s(L(F_3)) = 1$ and $s(L(F_2)) = 0$.

Theorem 1.4 yields an infinite family of counterexamples. The violation is, however, not confined to $s = 2$: in Section 5 we exhibit connected graphs with $s(L(G)) = 6$, and the data strongly suggest that $s(L(G))$ is unbounded.

The mechanism behind all of our examples is easy to describe. The signature is additive over connected components, and $L(C_5) = C_5$ has inertia $(3, 0, 2)$, so a disjoint union of k copies of C_5 has $s = k$; this is the disconnected counterexample noted in [1]. The content of Conjecture 1.1 is that connecting the copies destroys the surplus, and for the cheapest gluings this is exactly what happens: joining two 5-cycles by a single edge yields $s = 1$, which is the kayak paddle example of [1]. What we observe is that a suitably chosen *host* graph can absorb several 5-cycles while contributing enough positive eigenvalues of its own to leave a surplus behind.

2. A REFORMULATION

We use the standard relationship between the spectrum of a line graph and the signless Laplacian $Q(G) = D(G) + A(G)$, where $D(G)$ is the diagonal matrix of degrees; see [2, Chapter 1]. The following lemma restates Conjecture 1.1 in a form convenient both for computation and for the proof of Theorem 1.4.

Lemma 2.1. *Let G be a connected graph with $n \geq 2$ vertices and m edges, and let $q_1 \geq \dots \geq q_n$ be the eigenvalues of $Q(G)$. Put*

$$p = \#\{i : q_i > 2\}, \quad z = \#\{i : q_i = 2\}.$$

Then $\text{In}(L(G)) = (p, z, m - p - z)$, and consequently

$$s(L(G)) = 2p + z - m.$$

In particular, Conjecture 1.1 asserts that $2p + z \leq m + 1$.

Proof. Let B be the $n \times m$ vertex–edge incidence matrix of G . Then $B^\top B = A(L(G)) + 2I_m$ and $BB^\top = Q(G)$. The matrices $B^\top B$ and $BB^\top$ have the same nonzero eigenvalues with the same multiplicities. Write $r = \text{rank } B$. Since G is connected, $r = n$ if G is non-bipartite and $r = n - 1$ if G is bipartite; correspondingly $Q(G)$ has no zero eigenvalue in the first case and exactly one in the second. In both cases $Q(G)$ has exactly r nonzero eigenvalues, and $B^\top B$ has these together with $m - r$ zeros.

Hence the spectrum of $A(L(G)) = B^\top B - 2I_m$ consists of $q_i - 2$ over the r nonzero q_i , together with -2 with multiplicity $m - r$. Every q_i equal to 0 would likewise contribute -2 , so counting all n eigenvalues of $Q(G)$ uniformly we obtain $n^+(L(G)) = p$, $n^0(L(G)) = z$, and

$$n^-(L(G)) = \#\{i : 0 < q_i < 2\} + (m - r) = (r - p - z) + (m - r) = m - p - z. \quad \square$$

Lemma 2.1 makes the two regimes of Conjecture 1.1 transparent. Since $p + z \leq n$, we have $s(L(G)) \leq 2n - m$, so no counterexample can satisfy $m \geq 2n - 1$; this recovers the observation of [1]. Counterexamples must therefore be sparse, and this is where we look.

3. A COUNTEREXAMPLE ON FOURTEEN VERTICES

Proof of Theorem 1.3. Label the 4-cycle $a_1 a_2 a_3 a_4$, and let $b_1 \cdots b_5$ and $c_1 \cdots c_5$ be two 5-cycles. Let G_0 have edge set

$$\{a_i a_{i+1}\}_{i=1}^4 \cup \{b_i b_{i+1}\}_{i=1}^5 \cup \{c_i c_{i+1}\}_{i=1}^5 \cup \{a_1 b_1, a_2 c_1\},$$

indices modulo the respective cycle lengths. Then G_0 is connected, $n(G_0) = 14$ and $m(G_0) = 16$, and G_0 has ten vertices of degree 2 and four of degree 3.

The line graph $L(G_0)$ is a graph on 16 vertices whose adjacency matrix is an explicit integer matrix. Computing its inertia by symmetric Gaussian elimination over $\mathbb{Q}$ (Sylvester's law of inertia) gives $\text{In}(L(G_0)) = (9, 0, 7)$. Equivalently, by Lemma 2.1, the integer matrix $Q(G_0) - 2I$ has inertia $(9, 0, 5)$, so $p = 9$, $z = 0$ and $s(L(G_0)) = 2 \cdot 9 + 0 - 16 = 2$. $\square$

Since $9 > 7 + 1$, the graph G_0 refutes Conjecture 1.1. Note that $14 \leq 16 \leq 2 \cdot 14 - 2$, so G_0 lies in the open range of [1]; and since $n(G_0) = 14$ exceeds the exhaustive verification range of that paper, it is consistent with all computations reported there.

4. AN INFINITE FAMILY

Throughout this section fix $k \geq 1$ and let F_k be as in Theorem 1.4: a hub vertex u , together with k disjoint 5-cycles with vertex sets $\{v_{j,0}, \dots, v_{j,4}\}$ for $1 \leq j \leq k$, and the k edges $uv_{j,0}$. Then

$$n(F_k) = 5k + 1, \quad m(F_k) = 6k,$$

with $\deg u = k$, $\deg v_{j,0} = 3$ and $\deg v_{j,i} = 2$ for $i \neq 0$. Note $n \leq m \leq 2n - 2$ for all $k \geq 1$.

Write

$$M = \text{diag}(3, 2, 2, 2, 2) + A(C_5)$$

for the 5×5 block of $Q(F_k)$ supported on a single blade, where the first coordinate corresponds to the attachment vertex $v_{j,0}$.

Lemma 4.1. *The characteristic polynomial of M is $(x - 3)(x^2 - 3x + 1)(x^2 - 5x + 3)$; its eigenvalues are*

$$\frac{3-\sqrt{5}}{2}, \quad \frac{5-\sqrt{13}}{2}, \quad 3, \quad \frac{3+\sqrt{5}}{2}, \quad \frac{5+\sqrt{13}}{2}.$$

Consequently $\text{In}(M - 2I) = (3, 0, 2)$ and $\det(M - 2I) = 3$.

Proof. The stated factorisation is a direct computation. Numerically the roots are 0.38197, 0.69722, 3, 2.61803, 4 of which exactly three exceed 2 and none equals 2, giving $\text{In}(M - 2I) = (3, 0, 2)$. For the determinant, note $M - 2I = A(C_5) + e_0 e_0^\top$, where e_0 is the first standard basis vector, so by the rank-one determinant identity

$$\det(M - 2I) = \det A(C_5) + [\text{adj } A(C_5)]_{00} = \det A(C_5) + \det A(P_4) = 2 + 1 = 3,$$

since deleting $v_{j,0}$ from C_5 leaves a path on four vertices. $\square$

Proof of Theorem 1.4. The cyclic group $\mathbb{Z}_k$ acts on F_k by permuting the blades, fixing u . Let $\omega = e^{2\pi i/k}$. Decomposing $\mathbb{C}^{V(F_k)}$ into isotypic components for this action block-diagonalises $Q = Q(F_k)$ as follows.

For each nontrivial character, indexed by $j \in \{1, \dots, k-1\}$, the corresponding component consists of vectors whose restriction to blade t equals $\omega^{jt}y$ for a fixed $y \in \mathbb{C}^5$, and whose hub coordinate vanishes because $\sum_t \omega^{jt} = 0$. On such vectors Q acts as M . This accounts for $k-1$ copies of the spectrum of M .

The trivial character contributes a 6-dimensional component spanned by e_u together with the normalised symmetrisations of the blade coordinates. In this basis Q acts as

$$B_k = \begin{pmatrix} M & \sqrt{k} e_0 \\ \sqrt{k} e_0^\top & k \end{pmatrix},$$

the coupling $\sqrt{k}$ arising because the hub is adjacent to exactly one vertex of each blade. Since $5(k-1) + 6 = 5k + 1 = n$, the spectrum of Q is the union of $k-1$ copies of the spectrum of M and the spectrum of B_k .

It remains to count eigenvalues exceeding 2. By Lemma 4.1, each nontrivial block contributes exactly 3, and none contributes an eigenvalue equal to 2. For the trivial block, $M - 2I$ is nonsingular, so Haynsworth's inertia additivity formula applied to the Schur complement gives

$$\text{In}(B_k - 2I) = \text{In}(M - 2I) + \text{In}\left((k-2) - k e_0^\top (M - 2I)^{-1} e_0\right).$$

By Lemma 4.1 and Cramer's rule,

$$e_0^\top (M - 2I)^{-1} e_0 = \frac{[\text{adj}(M - 2I)]_{00}}{\det(M - 2I)} = \frac{\det A(P_4)}{3} = \frac{1}{3},$$

because deleting the first row and column of $M - 2I = A(C_5) + e_0 e_0^\top$ leaves $A(P_4)$. The Schur complement is therefore the scalar

$$(k-2) - \frac{k}{3} = \frac{2k-6}{3},$$

which is positive precisely when $k > 3$, zero when $k = 3$, and negative when $k < 3$.

Hence for $k \geq 4$ we get $\text{In}(B_k - 2I) = (4, 0, 2)$, and so, in the notation of Lemma 2.1,

$$p = 3(k-1) + 4 = 3k + 1, \quad z = 0.$$

Lemma 2.1 now gives $\text{In}(L(F_k)) = (3k + 1, 0, 6k - (3k + 1)) = (3k + 1, 0, 3k - 1)$ and

$$s(L(F_k)) = 2(3k + 1) - 6k = 2.$$

For $k = 3$ the Schur complement vanishes, so $\text{In}(B_3 - 2I) = (3, 1, 2)$, giving $p = 9$, $z = 1$ and $s = 2 \cdot 9 + 1 - 18 = 1$; for $k = 2$ it is negative, giving $p = 6$, $z = 0$ and $s = 12 - 12 = 0$. $\square$

Remark 4.2. The proof reproduces the three regimes exactly: F_2 satisfies Conjecture 1.1 strictly, F_3 attains equality, and F_k violates it for every $k \geq 4$. The transition is governed entirely by the sign of $(2k-6)/3$. We verified $\text{In}(L(F_k))$ independently by exact rational arithmetic for $2 \leq k \leq 12$; the values agree with the closed form above.

5. LARGER VIOLATIONS

Theorem 1.4 gives $s(L(G)) = 2$. Replacing the hub by a cycle produces larger signatures. For $h \geq 3$, let H_h be the graph obtained from a cycle C_h by attaching to *every* vertex a 5-cycle joined by a single edge, so that $n(H_h) = 6h$ and $m(H_h) = 7h$; again $n \leq m \leq 2n - 2$. Table 1 records the inertia of $L(H_h)$ for several values of h , computed by exact symmetric elimination over $\mathbb{Q}$.

h	n	m	p	z	$s(L(H_h))$
4	24	28	15	0	2
7	42	49	26	0	3
10	60	70	37	0	4
14	84	98	51	0	4
17	102	119	62	0	5
20	120	140	73	0	6

TABLE 1. Exactly computed inertia data for H_h . Here p and z are as in Lemma 2.1, so $\text{In}(L(H_h)) = (p, z, m - p - z)$ and $s = 2p + z - m$. In every case $z = 0$, so no eigenvalue of $Q(H_h)$ equals 2 and the classification is unambiguous.

In particular $s(L(G))$ is not bounded above by 2. Attaching two 5-cycles at each vertex of C_{11} gives a connected graph on 121 vertices with $s = 7$. The data are consistent with linear growth of $s(L(H_h))$ in h , subject to an arithmetic dependence on h that we have not analysed; we therefore record the following.

Conjecture 5.1. $s(L(G))$ is unbounded over connected graphs G . Indeed $s(L(H_h)) \rightarrow \infty$ as $h \rightarrow \infty$.

A proof should be within reach by the method of Theorem 1.4: the cyclic symmetry of H_h block-diagonalises $Q(H_h)$ into h blocks of size 6, one for each character $\theta_j = 2\pi j/h$, so that $p = \sum_j c(\theta_j)$ where $c(\theta)$ counts the eigenvalues of an explicit 6×6 matrix exceeding 2. Linear growth of $s = 2p - 7h$ then amounts to the statement that the mean of $2c(\theta)$ over $\theta \in [0, 2\pi)$ strictly exceeds 7.

6. CONCLUDING REMARKS

Conjecture 1.1 fails, but not by much on small graphs, and it remains natural to ask for the correct statement.

Proposition 6.1. *No connected graph on at most 12 vertices satisfies $s(L(G)) \geq 2$.*

Proof. By Lemma 2.1 and $p + z \leq n$ we have $s(L(G)) \leq 2n - m$, so any counterexample satisfies $m \leq 2n - 2$; and $s(L(T)) \leq -1$ for every tree T [1], so $m \geq n$. It therefore suffices to examine connected graphs with $n \leq m \leq 2n - 2$. Using **geng** [4] we generated all such graphs up to

isomorphism for $9 \leq n \leq 12$:

n	graphs	survivors
9	76,800	84
10	1,335,628	394
11	28,908,704	1,202
12	744,863,676	6,878
	775,184,808	8,558

For each graph the inertia of $Q(G) - 2I$ was computed by LDL^T decomposition with symmetric pivoting. Writing $a = \#\{q_i > 2\}$ and $b = \#\{q_i = 2\}$, we have $s(L(G)) = 2a + b - m \leq 2(a + b) - m$, so retaining every graph whose *upper* bound $2(a + b) - m$ is at least 2 cannot discard a counterexample. The survivors listed above were then resolved by exact symmetric elimination over $\mathbb{Q}$; every one satisfies $s(L(G)) \leq 1$. The case $n \leq 8$ is immediate from the same computation, and the case $n = 9$ independently reproduces the verification reported in [1]. $\square$

Problem 6.2. Determine the smallest order of a connected graph G with $s(L(G)) \geq 2$. By Proposition 6.1 it is at least 13, and by Theorem 1.3 at most 14; only the value 13 remains to be decided. The corresponding search space consists of approximately 4.9×10^9 graphs and is well within reach of a modest computation.

Problem 6.3. Find a sharp upper bound for $s(L(G))$ over connected graphs G in terms of natural parameters. Since s is additive over components and $s(L(C_5)) = 1$, a bound in terms of the number of vertex-disjoint induced 5-cycles, or more generally of induced subgraphs H with $s(L(H)) > 0$, would explain the constructions given here.

We remark that all of the above is consistent with the results proved in [1]: our graphs are neither trees nor dense, all satisfy $n \leq m \leq 2n - 2$, and each satisfies $n^+(L(G)) \leq 3n^-(L(G))$ as required by [1, Theorem 4.5]. Our computations also independently reproduce the tightness examples recorded there, namely $s(L(C_{4k+1})) = 1$ and $s = 1$ for two 5-cycles joined by an edge.

Computational remarks. All inertias reported here were computed twice: once in floating point from the eigenvalues of the relevant matrices, and once exactly, by symmetric Gaussian elimination over $\mathbb{Q}$ applied to the integer matrices $A(L(G))$ and $Q(G) - 2I$, invoking Sylvester's law of inertia. The two computations agree in every case, and in every counterexample reported here the exact computation returns $n^0 = 0$, so the sign classification does not depend on any tolerance. The exhaustive search underlying Proposition 6.1 was organised as a rigorous filter in floating point followed by exact rational resolution of the survivors, as described in its proof; only 8,558 of the 775,184,808 graphs examined required exact arithmetic.

Acknowledgement. The counterexamples reported here were found with the assistance of an AI system (Anthropic's Claude), which was used to design and run the computational searches and to verify the resulting inertias. All statements have been checked by exact rational arithmetic and the proofs are independently verifiable.

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